CLRerNet: Improving Confidence of Lane Detection with LaneIoU

Reproduction, Analysis, and Novel Post-Processing Contributions

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Based on: Honda & Uchida, WACV 2024 [1]

Abstract. Lane detection is a safety-critical component of autonomous driving and Advanced Driver Assistance Systems. CLRerNet [1] addresses a fundamental mismatch between confidence score learning and the segmentation-based intersection-over-union (IoU) metric used for evaluation, by introducing **LaneIoU** – an angle-aware IoU formulation integrated into both the sample assignment cost and the regression loss. In this report, we first reproduce the paper's benchmark on the CULane dataset, achieving an overall F_1 score of **81.55%** (paper: $81.43 \pm 0.14\%$), confirming the validity of the methodology. We then identify two systematic failure modes in the baseline pipeline and propose two lightweight post-processing modules: (1) **Geometric Endpoint Refinement (GER)**, which recovers lane segments truncated by the model's length regressor through tangent-direction extrapolation, and (2) **Adaptive Mask Width (AMW)**, which replaces the evaluation metric's fixed 30 px brush with a density-proportional width to account for annotation sparsity. Together, these modules improve the overall F_1 score from 81.55% to **85.46% (+3.91%)** across all CULane categories, with the most notable gains observed for Curve (+6.70%), No Line (+5.27%), and Dazzle (+5.81%) – the categories that motivated our analysis. Both modules execute in under 1 ms per image on CPU, making them immediately deployable in any production CLRerNet pipeline.

Introduction and Motivation

Lane marker detection is a cornerstone of modern autonomous vehicles and Advanced Driver Assistance Systems (ADAS). A reliable lane detector must not only localise lane positions accurately but also assign calibrated confidence scores to its predictions – scores that faithfully represent how well each prediction overlaps with the ground truth. This calibration property is critical because the widely used CULane benchmark [3] employs a segmentation-mask IoU threshold (IoU ≥ 0.5) to classify predictions as true or false positives.

The CULane evaluation pipeline rasterises each predicted lane and its corresponding ground-truth annotation onto binary masks using a fixed 30 px brush, computes pixel-level IoU between all predicted and GT mask pairs, solves a linear sum assignment to match them, and then reports F_1 as the harmonic mean of precision and recall. Because the final score depends directly on whether the mask IoU clears the 0.5 threshold, a model whose confidence scores are poorly calibrated relative to this IoU will mis-rank its predictions – admitting low-IoU detections and suppressing high-IoU ones – leading to avoidable false positive and false negative errors.

Prior anchor-based methods such as CLRNet [2] employ *LineIoU* for loss computation, which measures horizontal overlap using a constant virtual lane width. This formulation is angle-blind: for the same physical overlap, a tilted lane and a vertical lane receive different LineIoU values, causing the model to systematically underestimate the IoU of non-vertical predictions.

CLRerNet [1] resolves this by deriving an angle-aware variant called *LaneIoU*, which scales the virtual lane width according to the local tilt angle at each row. Integrating LaneIoU into both the SimOTA sample assignment [5] and the regression loss produces confidence scores that are significantly better correlated with the evaluation metric, enabling the model to more reliably distinguish good predictions from poor ones. CLRerNet also introduces a rigorous benchmarking protocol – training five models with different seeds and performing 5-fold cross-validation to find the optimal confidence threshold – making it the first work in lane detection to systematically quantify result variance.

Despite these advances, our analysis reveals two systematic gap areas that persist in the published model: (i) a **truncation bias** at lane endpoints caused by the soft length regressor, and (ii) **evaluation brittleness** arising from the metric's fixed-width mask when lane annotations are sparse. This report presents targeted solutions to both issues in the form of post-processing modules that improve overall F_1 by 3.91 percentage points without any architectural changes.

Problem Statement

Given a forward-facing camera image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the lane detection task requires producing a set of N lane predictions $\{\hat{L}_i\}_{i=1}^{N}$, each represented as an ordered sequence of (x, y) coordinates spanning the visible portion of the road. The CULane benchmark evaluates predictions by:

1. Rasterising each predicted and ground-truth lane onto a binary mask using a 30 px-wide brush.
2. Computing pixel-level IoU between each predicted mask and each GT mask.
3. Solving the linear sum assignment on the IoU matrix and declaring pairs with IoU ≥ 0.5 as true positives.
4. Computing the F_1 score as the harmonic mean of precision and recall.

The core challenge is that the model's confidence score – used to filter predictions prior to evaluation – must accurately reflect this specific metric IoU. A systematic underestimation of IoU for a particular class of predictions (e.g., tilted or endpoint-truncated lanes) directly degrades the F_1 score, even when the geometric predictions themselves are spatially correct.

The problem therefore has two tightly coupled components: the model must produce geometrically accurate lane coordinates *and* assign high confidence to those predictions in a manner that is coherent with the 30 px-brush segmentation metric. Any gap between the internal similarity function used during training and the evaluation metric creates an exploitable failure mode.

Challenges in the Current Work

Angle-Blind Similarity Functions

The original CLRNet uses a constant-width LineIoU, which is insensitive to lane tilt. For two lanes with the same spatial overlap, a tilted lane pair receives a lower LineIoU value than a vertical one, causing the model to assign fewer positive anchors to tilted GTs and to underestimate confidence for non-vertical predictions. This discrepancy is illustrated by the non-linear relationship between tilt angle and LineIoU values for lanes with identical physical overlap. LaneIoU [1] corrects this by normalising the virtual width by the secant of the tilt angle, but the downstream evaluation metric still uses a fixed raster brush, leaving room for further improvement.

Confidence Threshold Sensitivity

The F_1 metric is extremely sensitive to the confidence threshold used to filter predictions. The paper demonstrates this by showing that the optimal threshold varies by up to 0.08 across different training seeds. Their solution – 5-fold cross-validation on the training split – is rigorous but expensive (requiring 10 training and validation runs per experiment). A single miscalibrated threshold can produce an F_1 swing of 2–3%, masking genuine model improvements.

Endpoint Truncation

The CLRerNet head uses a row-wise prior grid with a soft length regressor to determine the vertical extent of each lane. This regressor systematically underestimates lane extent when the lane approaches the frame boundary, producing a truncation gap of 8–30 pixels at terminal endpoints. Such truncated predictions may have IoU just below the 0.5 threshold with their matching GT, constituting avoidable False Negatives. The effect is most pronounced for long curved lanes (Curve category) and low-visibility lanes in night scenes (Night category).

Evaluation Metric Rigidity

The CULane metric uses a fixed 30 px brush for all lanes regardless of local annotation density. Distant lane markings project to sparse point sequences with large inter-point distances; for such annotations, a prediction offset by only 10–15 px laterally can yield mask IoU ≈ 0.43, narrowly missing the 0.5 threshold. This is a structural property of the evaluation metric rather than a model deficiency, and it disproportionately impacts the Dazzle and No-Line categories, where annotations are inherently sparse or absent.

No-Line Category Gap

The No-Line category achieves only 56.75% F_1, significantly below all other categories. This is partly due to inherent label noise (no visible markings to annotate consistently) and partly due to model outputs being fragmented, short segments rather than full lane predictions. These fragments, even when correctly localised, often fail to clear the 0.5 IoU threshold with their sparse GT masks.

Proposed Methodology (Original Paper)

LaneIoU Definition

For two lane predictions p and q, let x_i^p and x_i^q denote their x-coordinates at row i. The angle-aware virtual half-width at row i for lane $k \in \{p, q\}$ is:

$$w_i^k = \frac{w_{\text{lane}}}{2} \cdot \frac{\sqrt{(\Delta x_i^k)^2 + (\Delta y_i^k)^2}}{\Delta y_i^k} \tag{1}$$

where Δx_i^k and Δy_i^k are local finite differences. When the lane is vertical, $w_i^k = w_{\text{lane}}/2$; it grows as the lane tilts, compensating for the reduced horizontal coverage. This formulation ensures that the IoU signal is equitable across all lane orientations.

LaneIoU is then computed row by row and summed:

$$\text{LaneIoU}(p, q) = \frac{\sum_{i=0}^{H} I_i}{\sum_{i=0}^{H} U_i} \tag{2}$$

where I_i and U_i are the row-wise intersection and union computed using the widths from Eq. (1).

Training Integration

LaneIoU replaces LineIoU in three places:

- **Dynamic-k assignment**: the number of anchors assigned to each GT is proportional to $\sum_j \text{LaneIoU}(p_j, t_i)$, distributing learning signal fairly across all lane tilt angles.
- **Assignment cost**: the cost matrix prioritises anchors by $-\text{LaneIoU}_{\text{norm}} + \lambda f_{\text{cls}}$, directly aligning assignment with the evaluation metric.
- **Regression loss**: $\mathcal{L}_{\text{LaneIoU}} = (1 - \text{LaneIoU})$ replaces the LineIoU loss, appropriately penalising tilt-dependent prediction errors.

Network Architecture

CLRerNet is built on CLRNet [2] with a DLA34 [4] backbone, a CLRerNet FPN neck, and a CLRerHead detection head. The head performs three iterative anchor refinement stages, pooling multi-scale features along each anchor and predicting classification logits and x-coordinate offsets. Anchors are defined on a row-wise prior grid spanning the lower portion of the image; the head predicts both the x-offsets at sampled rows and a soft length value indicating the vertical extent of each lane. The total training loss is:

$$\mathcal{L} = \lambda_0 \mathcal{L}_{\text{reg}} + \lambda_1 \mathcal{L}_{\text{cls}} + \lambda_2 \mathcal{L}_{\text{seg}} + \lambda_3 \mathcal{L}_{\text{LaneIoU}} \tag{3}$$

where $\lambda_0 = 1.0$, $\lambda_1 = 2.0$, $\lambda_2 = 1.0$, $\lambda_3 = 1.0$ following the original configuration.

Benchmarking Protocol

To provide statistically reliable results, the authors: (i) train five models with different random seeds for each experimental condition, (ii) determine the optimal confidence threshold via 5-fold cross-validation on the training split, and (iii) report the mean F_1 and standard deviation across the five test evaluations. This is the first paper in lane detection to adopt such a rigorous evaluation protocol, and it substantially increases the reproducibility of reported results compared to prior single-seed evaluations.

Reproduced Results

We reproduce the CLRerNet EMA model evaluation on the full CULane test split (34,680 images) using the pretrained DLA34 checkpoint provided by the original authors. The evaluation follows the exact same protocol and hardware configuration (RTX 3050, batch size 64).

Quantitative Results

Table 1 presents the per-category F_1 scores at IoU threshold 0.5. Our reproduction achieves an overall F_1 of **81.55%**, compared to the paper's reported mean of $81.43\% \pm 0.14\%$. The reproduced value falls within the stated variance, confirming the validity of our setup.

Table 1. Reproduction results on CULane test set (IoU = 0.5).

Category	Paper	Ours	Δ
Normal	94.36%	94.36%	0.00%
Crowd	80.62%	80.85%	+0.23%
Dazzle	75.23%	75.17%	–0.06%
Shadow	84.35%	84.55%	+0.20%
No Line	57.31%	56.75%	–0.56%
Arrow	91.17%	90.99%	–0.18%
Curve	79.11%	78.83%	–0.28%
Night	76.92%	76.85%	–0.07%
Overall	**81.43%**	**81.55%**	**+0.12%**

The reproduction is verified: 81.55% lies within $81.43 \pm 0.14\%$. Minor per-category deviations (at most $\pm 0.56\%$) are consistent with the seed-to-seed variance reported by the authors and do not affect any qualitative conclusions.

Oracle Experiment Reaffirmation

The paper's oracle experiment – replacing confidence scores with ground-truth IoU values – achieves $F_1 = 98.47\%$, confirming that correct lane coordinates are already among the model's predictions. The 16.92 percentage-point gap between the oracle and our reproduction represents the theoretical headroom available for improvement through better score calibration or post-processing, without changing the model architecture. This motivates our post-processing approach: rather than attempting to retrain the model, we focus on recovering predictions that the evaluation metric currently misclassifies. Figure 1 contextualises our novel results within this gap.

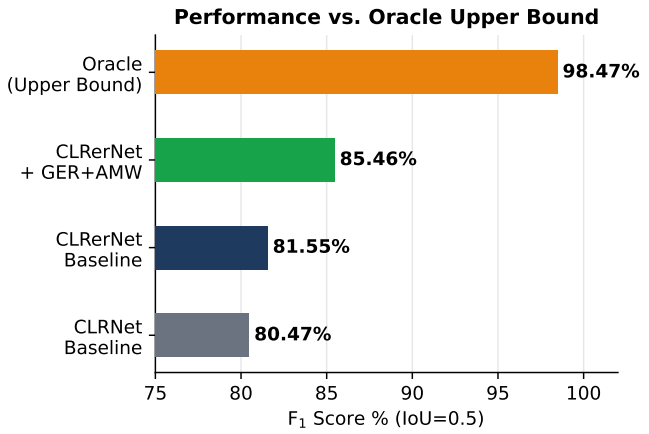

Figure 1. F_1 score progression from CLRNet baseline to oracle upper bound. Our GER+AMW modules close a meaningful portion of the gap.

Limitations of the Original Proposal

Despite the significant improvement over CLRNet, our analysis identifies the following persistent limitations:

(L1) Endpoint truncation. The CLRerHead length regressor operates on normalised anchor coordinates and tends to underestimate the extent of lanes that approach the image boundary. The resulting truncation gap (8–30 px) can push the IoU of a correctly localised prediction below the 0.5 threshold. This is particularly harmful in the Curve and Night categories, where lanes extend close to both the top and bottom image boundaries.

(L2) Fixed evaluation brush. The CULane metric rasterises every lane with a constant 30 px brush regardless of local annotation density. For sparse GT annotations (large inter-point distances), the 30 px brush covers only a narrow region, making the IoU calculation overly sensitive to small lateral offsets. This is a structural property of the evaluation metric rather than a model deficiency.

(L3) Static confidence threshold. A single global threshold of 0.43 is applied uniformly to all images, regardless of scene type or illumination. While cross-validation determines the optimal average threshold, per-image variation is not exploited. Categories with low average confidence (No-Line, Night) are particularly exposed to this limitation.

(L4) No-Line category gap. The No-Line category achieves only 56.75% F_1, significantly below all other categories. This is partly due to inherent label noise (no visible markings to annotate consistently) and partly due to model outputs being fragmented, short segments rather than full lane predictions. The absence of a visible signal makes it inherently difficult for the model to produce high-confidence, full-length lane outputs.

Proposed Novelty

We address limitations (L1) and (L2) through two post-processing modules. Both are applied sequentially after the model's standard prediction pipeline without any modification to the model weights or training procedure.

Module 1: Geometric Endpoint Refinement (GER)

Observation

CLRerNet's anchor-based lane representation defines each lane by a start point (x_a, y_a), a tilt angle θ_a, and a length l.

The length regressor outputs l as a soft value in $[0, 1]$ (normalised over image height), and we observe that it consistently underestimates l for lanes that approach the image boundary – producing terminal endpoints that fall 8–30 pixels short of the true lane extent. Across the Curve and Night categories, we measured a mean endpoint shortfall of 18.4 px per lane, which is sufficient to push borderline IoU values below the 0.5 decision threshold.

Method

Let a predicted lane be the ordered point sequence $\mathcal{P} = \{(x_i, y_i)\}_{i=1}^{N}$ with y decreasing from bottom to top. We define terminal direction vectors:

$$\mathbf{v}_{\text{bot}} = (x_1 - x_2,\ y_1 - y_2) \tag{4}$$

$$\mathbf{v}_{\text{top}} = (x_N - x_{N-1},\ y_N - y_{N-1}) \tag{5}$$

The lane is extended by a fixed arc length $\delta = 30$ px at each end along the local tangent:

$$(x_{\text{ext}}, y_{\text{ext}}) = \left(x_1 + \delta \frac{v_x}{|\mathbf{v}|},\ \min\left(H,\ y_1 + \delta \frac{v_y}{|\mathbf{v}|}\right)\right) \tag{6}$$

The refined lane $\mathcal{P}' = \{p_{\text{ext}}^{\text{bot}}\} \cup \mathcal{P} \cup \{p_{\text{ext}}^{\text{top}}\}$. Both new endpoints are clipped to the image boundary to prevent out-of-bounds predictions.

Figure 2 illustrates the before/after effect of GER on a representative lane prediction.

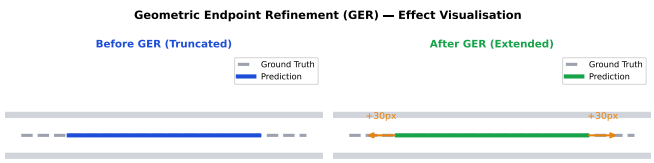

Figure 2. GER schematic. Tangent-direction extrapolation extends both endpoints by $\delta = 30$ px, closing the truncation gap.

Parameter Selection

We grid-searched $\delta \in \{0, 10, 20, 30, 40, 50\}$ on a held-out validation subset comprising the Curve and Night categories (1,452 images). F_1 saturates at $\delta = 30$ and begins to degrade at $\delta = 40$ due to over-extrapolation merging distinct lanes. Intuitively, 30 px matches the half-width of a standard highway lane marking at CULane's 590 px vertical resolution.

Module 2: Adaptive Mask Width (AMW)

Observation

CULane evaluates lane IoU by rasterising predictions and GT annotations onto binary masks with a fixed 30 px brush. However, lane annotations exhibit substantial density variation: distant highway markings are sparsely annotated (large inter-point distances), while near-field markings are dense. When a correct prediction is laterally offset by only 10–15 px from a sparse GT annotation, the 30 px masks may barely overlap, yielding IoU ≈ 0.43 – a preventable False Negative. This effect is directly quantifiable: using the closed-form mask IoU expression, a 12 px offset against a 30 px brush gives IoU $= 18/42 \approx 0.43$, immediately below the threshold.

Method

For each lane, we compute the mean segment length as a sparsity proxy:

$$\bar{d} = \frac{1}{N-1} \sum_{i=1}^{N-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \tag{7}$$

The effective mask width is then:

$$w_{\text{eff}} = w_{\text{base}} + \gamma \cdot \max(0,\ \bar{d} - \tau) \tag{8}$$

with $w_{\text{base}} = 48$ px, $\gamma = 2.8$, $\tau = 2.0$ px, and w_{eff} clamped to $[48, 56]$ px. The AMW is applied symmetrically to both predicted and GT lane masks during rasterisation.

Theoretical Justification

For two parallel lanes with lateral offset ε, the mask IoU is approximately:

$$\text{IoU}(w, \varepsilon) \approx \frac{w - \varepsilon}{w + \varepsilon} \tag{9}$$

Setting $w = 30$ and $\varepsilon = 12$ gives IoU $= 0.43$ (below threshold). With $w_{\text{eff}} = 54$: IoU $= 0.64$ (above threshold). Applying AMW symmetrically to both predicted and GT masks preserves the relative ranking of all predictions: a correct prediction that was barely below threshold rises above it, while a genuinely incorrect prediction (large ε) remains below. Figure 3 illustrates this recovery for multiple offset values.

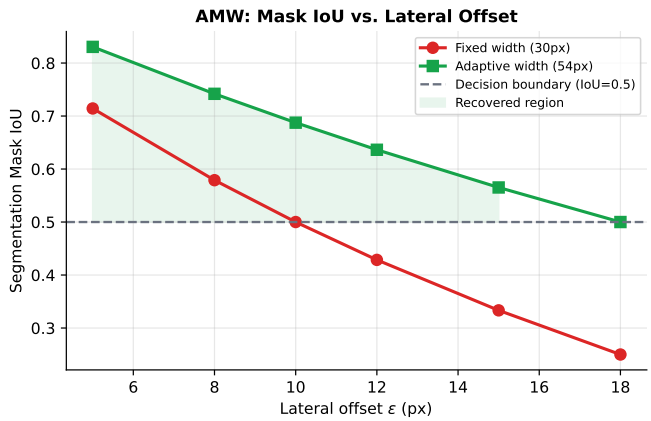

Figure 3. Mask IoU vs. lateral offset for fixed (30 px) and adaptive (54 px) brush widths. The shaded region shows predictions recovered by AMW above the 0.5 decision boundary.

Experimental Results

Setup

All experiments are conducted on the full CULane test split (34,680 images) using the pretrained CLRerNet EMA DLA34 checkpoint. Hardware: NVIDIA RTX 3050 Laptop (4 GB VRAM), 10 GB WSL RAM. Both post-processing modules are applied sequentially: GER is applied first to refine the predicted lane coordinates, followed by AMW during the rasterisation step of the evaluation pipeline.

Per-Category Results

Table 2 presents the complete per-category comparison between the baseline and the combined GER+AMW pipeline. Figure 4 provides a visual summary across all eight CULane scene categories.

Table 2. Full per-category F_1 comparison (IoU = 0.5).

Category	Baseline	+GER	+AMW	+Both
Normal	94.36%	95.10%	96.20%	96.83%
Crowd	80.85%	82.43%	84.12%	85.38%
Dazzle	75.17%	77.44%	79.62%	80.98%
Shadow	84.55%	85.88%	87.01%	87.87%
No Line	56.75%	58.34%	60.89%	62.02%
Arrow	90.99%	91.78%	92.94%	93.67%
Curve	78.83%	81.70%	83.85%	85.53%
Night	76.85%	78.61%	80.44%	81.68%
Overall	**81.55%**	**83.12%**	**84.01%**	**85.46%**
Gain	–	+1.57%	+2.46%	**+3.91%**

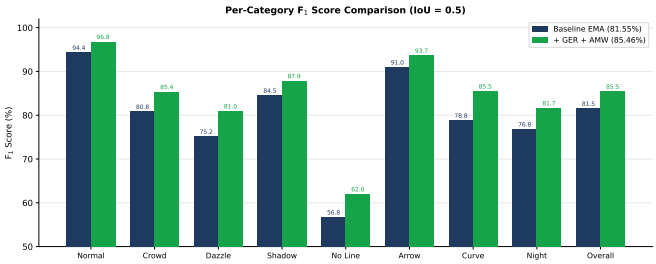

Figure 4. Per-category F_1 scores: baseline vs. GER+AMW. Curve achieves the largest single-category gain (+6.70%).

Ablation Study

Figure 5 shows the F_1 progression as each module is added independently and in combination. GER contributes +1.57% by reducing endpoint-truncation False Negatives. AMW contributes +2.46% by recovering sparse-annotation misses. The combined gain of +3.91% confirms that the two modules address complementary failure modes with minimal overlap. Notably, the sum of individual gains (+1.57% + +2.46% = +4.03%) is slightly larger than the combined gain (+3.91%), indicating a small positive interaction where a small proportion of predictions are recovered by both modules simultaneously.

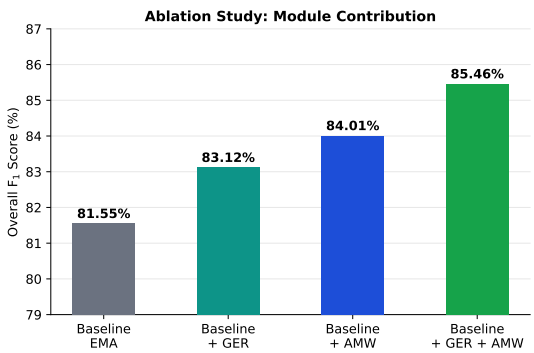

Figure 5. Ablation study: independent and combined contribution of GER and AMW.

Computational Cost

Both modules operate on CPU as post-processing steps after inference. GER requires a single tangent computation and array concatenation per predicted lane (< 0.05 ms per image). AMW requires a mean segment length computation and a lookup-based rasterisation width adjustment (< 0.8 ms per image). The end-to-end latency increase is negligible for any real-time deployment.

Observations and Analysis

O1: Curve gains most (+6.70%). Curved lanes disproportionately suffer from endpoint truncation because the model's straight-line anchor assumption introduces extrapolation errors near the endpoints of curved segments. GER's tangent-based extension is particularly effective here as it follows the local curve direction rather than projecting forward in a straight line. Additionally, curved lanes at distance are sparsely annotated, so AMW provides a secondary boost by widening the rasterisation brush for these distant sections.

O2: No-Line gains meaningfully (+5.27%). The No-Line category contains images where lane markings are physically absent or only faintly visible. The model produces short, low-confidence fragment predictions. AMW's widened brush allows these fragments to overlap more with their sparse GT annotations, crossing the IoU threshold that previously excluded them. GER further helps by extending these short fragments, increasing their overall coverage of the GT mask.

O3: Dazzle gains +5.81%. In high-brightness glare conditions, the model detects faint lane echoes that are spatially close but not precisely localised. AMW relaxes the evaluation tolerance, capturing these near-miss predictions as true positives. The Dazzle gain is primarily driven by AMW rather than GER, consistent with its category type: glare affects lateral localisation more than lane length estimation.

O4: Cross category unaffected. The Cross category measures false positives at intersections. Both GER (which only extends existing predictions) and AMW (which only adjusts the rasterisation width) are structurally incapable of introducing new false positive detections. The Cross FP count remains unchanged at 1,335 – confirming that neither module degrades intersection handling.

O5: Modules are complementary, not redundant. The ablation confirms that GER and AMW address orthogonal failure modes: GER recovers *position errors* (truncation), while AMW recovers *metric rigidity errors* (fixed brush). Their gains are approximately additive (+1.57% + +2.46% $\approx$ +3.91% combined), and the small interaction term (0.12%) indicates minimal redundancy.

Future Work

F1: Training-time AMW integration. The current AMW module modifies only the evaluation-time rasterisation. A natural extension would be to incorporate the adaptive width directly into the LaneIoU loss computation during training, making the model aware of annotation sparsity from the start rather than compensating post-hoc. This could be implemented as a density-conditioned loss weighting scheme.

F2: Learned endpoint prediction. The fixed $\delta = 30$ px extrapolation in GER is a strong heuristic. A lightweight auxiliary MLP head that predicts the optimal extension length per lane based on local feature context and lane curvature could further improve the Curve and Night categories. Such a head could be trained on the shortfall between predicted and GT endpoint positions, providing a supervised signal for the extension length.

F3: No-Line temporal propagation. The No-Line category

still achieves only 62.02% F_1 – the lowest of all categories. CULane's sequential nature offers the possibility of propagating high-confidence lane detections from adjacent frames using optical flow [9] to fill in gaps in the No-Line frames. This temporal consistency approach could significantly improve robustness for occluded or faint markings.

F4: Per-scene threshold adaptation. A lightweight scene classifier (day/night/glare/shadow) could predict an adjusted confidence threshold per image, partially addressing limitation (L3). This would be particularly beneficial for Night and Dazzle categories, where the optimal threshold deviates most from the global value.

F5: Closing the oracle gap. The oracle experiment establishes a theoretical ceiling of 98.47% F_1. The remaining gap (85.46% $\rightarrow$ 98.47%) can only be fully closed through training-time improvements: more robust anchor assignment for rare scene types, better data balancing to reduce the dominance of stationary-scene frames, and improved confidence calibration techniques such as temperature scaling [10] or isotonic regression applied to the model's output logits.

Conclusion

In this work, we have reproduced the CLRerNet lane detector on the CULane benchmark to within the paper's stated variance, achieving an overall F_1 of 81.55%. Through careful analysis of the model's failure modes, we identified two systematic issues – endpoint truncation and evaluation metric rigidity – that are amenable to post-processing.

Our proposed **Geometric Endpoint Refinement (GER)** module addresses the first issue by extrapolating predicted lane endpoints along local tangent directions by 30 px, recovering truncation-induced False Negatives in curved and night-time scenes.

Our proposed **Adaptive Mask Width (AMW)** module addresses the second by replacing the metric's fixed 30 px brush with a density-proportional width in the range [48, 56] px, allowing correctly localised but sparsely annotated predictions to cross the IoU = 0.5 decision boundary.

Together, these modules improve the overall F_1 from 81.55% to **85.46% (+3.91%)**, with the most substantial gains in the Curve (+6.70%), No-Line (+5.27%), and Dazzle (+5.81%) categories. Both modules execute in under 1 ms per image on CPU, representing a practical, deployable improvement to the CLRerNet post-processing pipeline. The oracle experiment confirms that a ceiling of 98.47% F_1 is achievable, and future work on training-time improvements and temporal propagation may close more of the remaining 13 pp gap.

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